

ERP1

Usage Guide

Batch manufacturing ERP/MES — orders, recipes, inventory, quality, shipping, purchasing, planning and accounting in one system.

July 10, 2026

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Welcome to ERP1

ERP1 is the plant's manufacturing and business system — the in-house replacement for the legacy Mar-Kov CMS. It runs in your web browser and covers the full life of a product: recipes, production batches, quality testing, inventory and lot tracing, shipping and invoicing, purchasing and receiving, planning, and accounting export.

How the system is organized

Everything is reached from the left-hand navigation. The pages fall into eight groups, and this guide has a chapter for each:

1. **Getting started & administration** — signing in, passwords and MFA, users, roles, permissions, electronic-signature settings, configuration, the audit trail, and the legacy import.
2. **Master data** — entities (customers, suppliers, warehouses), items, units, and QC test definitions.
3. **Recipes** — authoring, versioning, publishing and costing the master formulas.
4. **Production orders** — creating, executing, completing and (when needed) reversing batches and packouts.
5. **Sales & shipping** — shipping orders, staging assemblies, recording shipments, reversing a shipment, packing slips and invoices.
6. **Purchasing & receiving** — purchase orders, supplier pricing, receipts and supplier bills.
7. **Inventory & quality** — on-hand stock, adjustments, transfers, count sheets, lot tracking, recall tracing, sample sets, dispositions and certificates.
8. **Planning, viewers & accounting** — the material plan, analysis viewers, e-mail notifications, GL/tax setup and the accounting export.

Permissions, signatures and approvals

Three mechanisms control who can do what. It helps to recognize them on sight:

- **Programs** are page/action permissions bundled into **roles** (Administration → Roles). If a page or button is missing for you, your roles lack its program — ask an administrator.
- **Secured items** put an electronic signature on regulated actions (publishing a recipe, completing or reversing an order, QA dispositions). Depending on configuration, the signing dialog asks for a reason, your password, your MFA code, and sometimes a second person's (witness) credentials. If your group is not permitted to perform the action at all, a supervisor can authorize it on the spot by entering their own credentials — the supervisor's name goes on the signature record, acting on your behalf.
- **Approval policies** route certain edits (order edits, line edits, QA dispositions) through a request-and-approve queue when your role is request-only.

Every change in ERP1 is recorded in a tamper-evident audit trail, and reversals never erase history — they post compensating records.

A note on the legacy system

During parallel running, ERP1 mirrors the legacy CMS database. The mirror updates **only when an operator runs an import** from Administration → Legacy Import — nothing happens automatically, so the plant controls exactly when data is pulled. Records imported from the legacy system are generally read-only history in ERP1; new work is created natively.

Getting started & administration

This section covers signing in, managing your own account security, and the administrator screens that control who can use ERP1 and what they may do. The account, role, and security screens (Users, Roles, Secured Items, Approval Policies, Configuration, Audit, Import) each require the matching program to be granted to your group — if a link is missing from the top navigation bar, ask an administrator.

Signing in

1. Open ERP1. You land on the **Sign in to ERP1** card.
2. Enter your **Email** and **Password**, then click **Sign in**.
3. If two-factor authentication (2FA/MFA) is on for your account, the password step is accepted and an **Authenticator code** field appears with the note "Two-factor authentication is on for this account." Type the 6-digit code from your authenticator app and click **Verify code**.
4. Lost your phone? Click **Lost the authenticator? Use a recovery code**, enter one of your saved recovery codes (format xxxxx-xxxxx), and verify. **Use the authenticator app instead** switches back.
5. To start over with a different login, click ← **Different account**.

A wrong password or wrong code shows a red error under the form. If your account is disabled, sign-in is refused — ask an administrator to re-enable it.

Changing your password (and the forced first change)

New accounts (and any account an administrator reset) are flagged to change their password. On first sign-in you are sent to a full-screen **Set a new password** card ("You must change your password before continuing") — you cannot reach the rest of the app until you do. Enter your current password and a **new password of at least 12 characters**, then click **Update password**.

To change your password any time, click **your name** at the top-right of the header, which opens the **Change password** screen. The same 12-character minimum applies.

Setting up two-factor authentication (MFA)

On the **Change password** screen (below the password card) is the **Two-factor authentication** card, where you manage MFA yourself:

1. Type **your password** and click **Set up authenticator**.
2. Scan the QR code with an authenticator app (Google Authenticator, Microsoft Authenticator, Authy, etc.), or type the shown secret manually.
3. Enter the **6-digit code** the app shows and click **Confirm & enable**.
4. ERP1 shows your **recovery codes once** — store them somewhere safe. They are your way back in if you lose the authenticator.

Once enabled, the card reads "On — an authenticator code is required at sign-in and for electronic signatures" and shows how many recovery codes remain. **Re-enroll** (new device) needs your password plus a current code; **Turn off** needs your password plus a current code or a recovery code.

Signing in with single sign-on (SSO)

If your organization has configured SSO, a button with your provider's label appears below the sign-in form. Click it to authenticate with your identity provider (for example, your company Microsoft account) and you are returned signed in. SSO-only accounts have no ERP1 password and no ERP1-managed MFA — those are handled by your identity provider. If such an account needs to electronically sign records, an administrator must set a password for it (see below).

Users

The **Users** screen is the administrator's account desk.

- **Add user:** fill in **Email** and **Display name**. Leave **Initial password** blank to create an SSO-only account, or set one (min 12 characters). Optionally set the **SSO subject** (the identity provider's sub claim; for Entra ID this is the user's object id) and a starting **Role code**. Click **Create user**.
- The table shows each user's roles, **Sign-in** badges (**password**, **MFA**, **SSO**), and **Status** (ACTIVE / DISABLED).
- Row actions: **Edit roles** (tick the groups they belong to, **Save groups**); **SSO** (link or unlink their identity-provider subject); **Set/Reset password**; **Reset MFA** (only shown if they have MFA — they will then sign in with password only until they re-enroll); and **Disable/Enable**.

When an administrator sets or resets a password, the user must change it at their next password sign-in. Setting a password is also how you give an SSO-only user the ability to electronically sign records.

Roles (programs = permissions)

A **role** is a user group, and the **programs** ticked for that group are the screens its members may open — this is how permissions work. Click **New role** to create one (Code such as QA_MANAGER, Name, optional Description). Click **Manage** on a role to tick the programs it needs (grouped by folder) and **Save programs**; you can also edit its name/description or **Delete role**. Roles marked **system** are maintained by ERP1 and their programs cannot be edited here. Approval capabilities are set separately on the Approval Policies screen.

Secured Items

Secured Items define, per sensitive action, what sign-off ERP1 demands. For each action you set the **Response level** — any of **Require reason**, **Require signature**, **Require witness**, or **Disabled** — then, under **Group grants**, tick which groups **may perform** the action and which **may witness** it. Save the two sections independently. A **disabled** item fails safe: its action still requires a signature.

These settings drive the electronic-signature prompt users see when they perform a controlled action (closing an order, reversing a shipment, releasing QA, and so on). Depending on the item, that prompt asks for: a **reason**; the signer's **password** and, if they have MFA, an **MFA code**; a **witness** (a second person's email, password and MFA plus an optional note); and, if the operator's group lacks the perform grant, a **supervisor elevation** (a supervisor's email, password and MFA — that supervisor's signature goes on the record).

Approval Policies

This screen assigns approval capabilities (Request approval, Approve, Approve update, Approve change, Override, No approval required) to each group. Tick the boxes and click **Save** per row; a group with no saved policy shows its defaults (marked "default"). Note the on-screen caveat: these policies are configuration for the workflow engine and are **not yet enforced** on a specific action.

Configuration

Configuration holds application settings, arranged in tabs like the legacy Configuration Update screen. Pick a tab, change the fields (text, number, on/off toggles, dropdowns, passwords, or an image), and click **Save changes** — changes apply immediately, no restart. To set the **company logo** (used on printed documents), use the image setting: choose a PNG, JPEG, GIF or WebP file under ~300 KB; a preview appears with a **Remove** option. Oversized files are refused with "Logo too large." Some settings are read-only and shown greyed out.

Audit trail

The **Audit** screen is an append-only, hash-chained log of every change: **When**, **Actor**, **Action**, and **Summary**, and you can click a row to expand the exact field-level before→after values. Click **Verify integrity** to re-walk the chain; it reports "✓ Chain intact" with a record count, or flags exactly where any tampering occurred.

Legacy Import

The **Import** screen (titled "Legacy import") copies data **read-only** from the legacy Mar-Kov CMS into ERP1. It has two buttons, and — importantly — **nothing imports automatically. Both actions run only when an operator clicks them, so the plant controls the import window.**

- **Run full import** copies every mirrored table and sets the change-feed watermark. It is idempotent — re-running it is safe.
- **Sync changes** then pulls only what changed in legacy since the last import or sync (driven by the legacy audit log), run it whenever you want the mirror caught up during the transition. If a sync cannot apply every change it **holds the watermark and reports "failed"** — resolve the cause and run it again.

Use **Compare with legacy** (Reconciliation) to see the current watermark, how many operations are not yet synced, and per-table row-count drift between legacy, the ERP1 mirror, and native ERP1 data. **Recent runs** lists each run's mode and status (success/failed, with the error). The database connection itself is set up by an administrator outside the app.

Master data: entities, items, units, tests

Master data is the shared reference information the rest of ERP1 draws on: who you buy from and sell to, what you make and stock, the units you measure it in, and the quality tests each item must pass. You reach these screens from the top navigation bar: **Entities**, **Items**, **Units**, **Item Tests**, and **Lot Tracking**. Each list has a search box, sortable columns, and an **Export** button that downloads the current view.

Entities (customers, suppliers, warehouses, carriers)

An "entity" is any outside or internal party — a customer, a supplier, a warehouse, a freight carrier, a lab, a salesman. One entity can wear several hats at once, controlled by role checkboxes.

To add one:

1. Click **New entity**.
2. Enter an **Entity code** (required, up to 20 characters) and an optional **Name**.
3. Tick the **Roles** that apply: Supplier, Mfr (manufacturer), Customer, Ship-To, Salesman, Carrier, Warehouse, Lab.
4. If you tick **Ship-To**, a **Parent customer** search box appears — use it to attach this delivery address to the customer it belongs to (optional).
5. Click **Create entity**.

To change an existing entity, click **Edit** on its row. A pop-up opens with two areas. The top form holds the **Roles**, plus **Currency**, **Terms**, and **Customer type**, and an **Inactive (hidden from pickers)** checkbox that keeps old records intact while removing the entity from search lists. If Ship-To is ticked you can set the **Parent customer** here too. Click **Save changes**.

The lower area is the **address book**, which feeds the "To" and "Ship-To" blocks printed on invoices, packing slips and purchase orders. Each address has a **Type**:

- **Primary (document) address** — the main mailing address shown on documents. An entity can have only one.
- **Ship-to address** — a delivery location.

Click **+ Add address**, choose the type, fill in name, street lines, city, state, zip, country, contact, phone and email, then **Add address**. Use **edit** or **remove** on any existing address.

Common refusals: "Entity code already exists" (codes are unique); "A primary address already exists; edit it instead of adding another"; "An entity cannot be its own parent." Editing entities requires the Entities program — ask an administrator if the menu is missing.

Items (materials, products, names, packages)

The Items list covers everything you stock or make. Filter it by **type** — Materials, Packaged Products, Names, Prototypes, Packages — and by **Controlled only** for controlled substances (shown with an amber "Controlled" badge).

To create an item:

1. Click **New item**.

2. Enter an **Item code** (required), **Description**, **Unit** (for example kg), and pick a **Type**.
3. If the type is **Names**, an **Alias of** search box appears and is required — a name is just a label that points at a real stock item, so choose its target. The button stays disabled until you do.
4. Click **Create item**.

Click **Edit** on a row to open the item pop-up, which has three sections:

- **Details** — Description, Unit, Status, Specific gravity, Retest period (days), and checkboxes for Controlled substance, Certified organic, and No expiry. Names also show their **Alias of** target here. Click **Save details**.
- **Planning** — the knobs the planning engine reads: **Minimum stock**, **Lead time (days)**, and **Testing lead (days)**. Click **Save planning**.
- **Packaged products** — bindings that make a bulk item orderable as a finished pack. Pick a **Packaging prototype** and a **Packaged product**, then **Add binding**. Existing bindings list underneath and are marked "inactive" when retired.

Common refusals: "Item code already exists"; "An item cannot alias itself"; "That packaged-product binding already exists for this item"; and, on planning, a message about not being able to determine the site owner — that means the plant's site record isn't set up as expected; raise it with an administrator. Items management requires the Items program.

Units

The Units screen lists the units of measure the system uses (code, description, category, and any base-unit conversion). To add one, click **New unit**, fill in **Code** and **Description** (both required), and optionally a **Category** (such as WEIGHT), a **Base unit**, and a **Base qty** that expresses how this unit converts to the base. Click **Create unit**. Existing units are view-only here; the refusal "Unit already exists" means the code is taken. This screen needs the Units program.

Item Tests (per-item QC specs and the test catalog)

This screen shows the quality-control tests an item must pass — the same requirements that drive an order's quality section and its Certificate of Analysis.

1. Search for an item by code or description and select it. The result shows how many tests each item already has.
2. The **Test requirements** table lists each test with its Specification, Target, Group, Grade, and **Stages**.
3. Click **+ Add test**, or **Edit** on a row. Fill in the **Test name** (start typing to pick from the catalog), **Group**, **Grade**, numeric **Min / Max / Target**, an optional free-text **Specification** (which overrides min/max when set), **Qualifier**, and **Comment**.
4. Tick the **Stages** the test applies at: Receipt, Production, Retest. Save with **Add test** or **Save changes**. Use **remove** to delete a requirement.

At the bottom, the collapsible **Test catalog** is the master list of test names the picker offers. Expand it to add or edit catalog entries: **Test name**, **Description**, **Result type** (NUM numeric or BOOL pass/fail), **Precision** (numeric tests only), **Group** (required), and **Unit**. A test that is in use shows a "used by" count and cannot be removed until those requirements are cleared; test names can never be renamed, because

item requirements, order specs and sample results all link by name.

Viewing item tests needs the Item Tests program. Editing requirements needs the **Edit Item Test Requirements** right, and editing the catalog needs the **Edit Test Catalog** right.

Enabling lot tracking on an item

Lot tracking is switched on per item from the **Lot Tracking** screen (not the Items screen). Until an item is enabled it is consumed FIFO by quantity and shows a grey "FIFO (not traced)" badge; once enabled it shows a green "Lot-traced" badge.

1. Find the item and click **Enable....**
2. Pick a **Location / warehouse**, then add one row per lot: choose **Raw (vendor lot)** — where you enter the vendor/manufacturer lot and ERP1 assigns its own number — or **Finished good**, where you enter an existing lot number. Fill in **Qty on hand** and optional **Unit cost**.
3. Use **+ add lot** and **+ add another location** to enter all opening stock, then click **Enable lot tracking**.
4. ERP1 shows a table of the lot numbers it minted — print or label your physical stock with these.

Read the amber warning first: **enabling replaces the item's current on-hand inventory with the lots you enter**, so count carefully. The system refuses to enable if the item has an undispositioned QC sample from a native batch completion, or stock staged to a shipping order — disposition, reverse, ship or unstage first. It also rejects a lot number that already belongs to a different item, or the same lot entered twice for one location. **Disable** on a lot-traced item reverts it to FIFO by quantity. This screen needs the Lot Tracking program.

Recipes

The **Recipes** screen (top navigation: **Recipes**) is where you build, revise, cost, and release the formulas that production orders run from. A recipe is either a **Batching** recipe (makes a bulk product) or a **Packaging** recipe (makes a finished, packaged good). Every recipe is a *draft* until you publish it, and only published recipes can be ordered.

Browsing recipes

The list shows one row per recipe with **Recipe #**, **Type** (Batching or Packaging), **Status**, **Revision note**, and **Updated** date. Status is a colored badge: grey **Draft**, green **Active**, amber **Inactive**, plus a purple **Rework** tag where it applies.

1. Type in the search box to find a recipe by number.
2. Use the first dropdown to filter by **Type** (All types / Batching / Packaging).
3. Use the second dropdown to filter by state (All states / Active / Inactive / Draft).
4. Click a column header (Recipe # or Updated) to sort. Click **Export** to download the current list.
5. Click **View** on any row to open its detail card below the list.

Creating a draft

1. Click **New recipe**. A "New draft recipe" panel opens.
2. Choose **Type**: *Batching (bulk product)* or *Packaging (finished goods)*. Pick this carefully — type cannot be changed later.
3. Enter a **Recipe #** (up to 20 characters, e.g. UV2905.01). Numbers must be unique.
4. In **Product (item made)**, start typing an item code or description and pick the item the recipe produces.
5. Optionally enter a **Reference**.
6. Enter a **Recipe comment** — this is the revision note and is **required** (e.g. NEW).
7. Click **Create draft**. The draft opens for editing. It is not orderable until you publish it.

Refusals: the button stays disabled until a number, product, and comment are all filled in. "Recipe number ... already exists" means that number is taken (matching ignores upper/lower case).

Editing the header and procedure

Open the draft and click **Edit draft** (this button only appears while the recipe is still a draft — published recipes are read-only). The editor has two parts.

Header — edit the **Revision note (comment)**, **Reference**, **Lead time (days)**, and **Product**, and tick **Rework recipe** if this formula should be exempt from the single-active rule (see below). Click **Save header**. The note and a product are required.

Procedure — the ordered list of ingredient and instruction lines. Quantities are always authored **per 100 lb of product** (the formula basis), so they read as percentages of the batch. A running total, "Ingredients: X / 100 lb," turns green when your ingredients add up to 100.

1. Click **Add ingredient**, search, and pick an item. It appears as a new row; type its quantity (per 100 lb) in the right-hand box.
2. Click in **Add instruction**, type the step (e.g. MIX WELL), and press Enter or **Add**. Instruction text is shown in capitals.
3. Use the ↑↓ arrows to reorder any line, and **remove** to delete one.
4. Click **Save procedure**.

Refusals: Save is blocked until every ingredient has an item and a quantity greater than zero, and every instruction has text. "Recipe ... is published and immutable" means someone published it while you were editing — reload and clone instead.

Cloning to the next revision

You cannot edit a published recipe. To revise one, open it and click **Clone** → **new version**. This copies every line into a fresh draft and auto-numbers it using the .NN convention — cloning UV2905 produces UV2905.01, then .02, and so on. Only Batching and Packaging recipes can be cloned. Edit the new draft, then publish it; publishing the new revision automatically deactivates the old one (the **single-active rule**: at most one active recipe per product, unless it is marked Rework).

Verifying and publishing

Publishing is the controlled step that makes a recipe orderable. Click **Publish...** on a draft. The system first **verifies** the recipe and refuses if any check fails ("Recipe failed verification: ..."). Verify checks that: there is a product line, the product line has an item and a quantity greater than zero, there is at least one ingredient, no ingredient line is missing its item or quantity, and the revision note is present.

The Publish panel then asks for whatever your site requires (this is governed by the **recipe.publish** program; ask an administrator if you can't publish):

- **Reason / revision note** — sometimes optional, sometimes required.
- **Your password (e-signature)** — confirms it is really you.
- **MFA code** — if you are enrolled in multi-factor authentication.
- **Witness** — a second, different qualified user signs with their own email, password, and MFA code.
- **Supervisor authorization** — if your group is *not* permitted to publish, a supervisor can authorize it in place by entering their email, password, and MFA; they become the signer and you are recorded as acting on their behalf.

Click **Publish**. On success you'll see "Published ... — deactivated ..." naming any prior revision that was switched off.

Activate / deactivate

Published recipes carry an **Activate/Deactivate** button. Deactivating takes a recipe out of use without deleting it. Re-activating a non-rework recipe automatically deactivates whichever sibling revision was active, so there's never more than one active recipe for the same product. Drafts show a **Delete draft** button instead (with a confirmation prompt); published recipes can never be deleted, only deactivated.

Batch-record preview (rescale)

Click **Preview batch record** to see exactly how the batch ticket will look in production. A yellow banner reminds you this is **preview only — no batch number or lot is assigned** (those come from a real batching order). Enter a **Batch size** and click **Rescale** to see every ingredient quantity recomputed for that batch.

Expected cost (pricing panel)

Click **Expected cost** to price a batch. Enter a batch size and click **Recalculate**. Each ingredient row shows the quantity **Needed**, the cost **Supplier**, **Unit price**, **Order qty**, **Cost**, and **Excess**. The engine picks the cheapest supplier across that supplier's quantity-break tiers; if a tier's minimum order exceeds what you need, the surplus shows as **Excess**. Ingredients you make in-house (no supplier price) roll up from their own **active costing recipe** — shown with a blue sub-recipe badge — priced all-or-nothing. If neither is available, it falls back to the item's **standard cost**, then its **replacement cost**; anything with no price at all is flagged amber as **unpriced** and counted under the total. The footer shows **Total expected cost** and total excess.

*(The Recipes screen also has a **Replace ingredient** tool for swapping one item for another across many active recipes at once — it creates a .NN revision of each and can publish them in a batch, using the same e-signature rules as a single publish.)*

Production orders (batches & packouts)

Production work lives under **Orders** in the top navigation. A production order is either a **Batch** order (make bulk product from a recipe) or a **Packaging** order (pack bulk into finished units). This section covers creating, printing, executing, completing, revising, and reversing them.

The Orders browser

The Orders page lists every order — purchase, batch, packaging, and shipping — newest first.

1. Use the **search box** to find an order by number or party.
2. Use the **type dropdown** (All types / Batching / Packaging / Purchasing / Shipping) to narrow the list.
3. Tick **Open only** to hide completed and closed orders.
4. Click any column header (Order #, Type, Status, Ordered, Completed) to sort.

Each row shows the order number, type, customer/party, reference, **Status**, any **Hold** (an amber badge — user hold, execution hold, or Credit), the ordered date, and the completed date (or "open"). Statuses read **Not started**, **Released**, **Being edited**, **Completed**, or **Closed**. Click **View** to open the order detail card below the list.

Creating a batch order from a recipe

1. Click **New order** (top right).
2. In **Recipe**, start typing a recipe number. Matching production recipes appear, tagged Batch or Packaging. The list also surfaces a **Packaged products** group — picking one selects its packaging recipe instead.
3. Enter the **Batch size** (the field reads **Packaging quantity** for a packaging recipe). All line quantities scale from this.
4. Optionally set a **Required date** and a short **Reference** (up to 20 characters).
5. Click **Create order**.

The order is created **Not started**, its type following the recipe (batch or packaging), and it is assigned its manufacturing lot number (shown as **Mfg Lot**). Batch orders also pull the product's QC test specifications. If the recipe has no active lines you will see "Recipe has no active detail lines" — the recipe needs a published, active version first.

Before releasing, you can click **Edit order** to rescale the batch size or change the reference/required date. If your group only has request rights, the edit is **submitted for approval** and takes effect once a qualified approver approves it.

The batch sheet printout

From an open batch or packaging order, click **Print batch sheet**. A printable ticket opens in a new tab — **Batch Ticket** for a batch order, **Packaging Record** for packaging. It auto-fills the formula number, dates, product, total weight, **This Lot #** and **Last Lot #**, and the full **Procedure** (raw materials with grams and pounds, plus instruction rows). The **Batch Additions**, **Quality Control** results, **Packaging**, and **sign-off** cells (QC'd by, Weighed by, Mixed by, Packed by, Closed by) are left blank for hand-recording on the floor.

Click **Print** to send it to the printer.

Releasing the order

On a Not-started order, click **Release** in the Lifecycle bar. The status becomes **Released**, and the guided-execution tools unlock. If someone already released it you'll see "Cannot release order #N: it is Released (must be Not started)."

Guided execution — recording actuals

On a Released order, click **Guided execution (dispense / record actuals)**.

1. For each **material line**, the planned quantity is shown. Enter the **actual quantity** dispensed (it defaults to the planned amount) and click **Record**.
2. For a **lot-traced** item, lot rows appear after you enter a quantity. Click an on-hand lot chip (each shows the lot and quantity on hand) or type a lot, and enter how much came from each. The lot quantities must sum to the actual, or the line won't record.
3. For **instruction** lines, click **Done**.
4. To add an ingredient not in the recipe, click **+ Batch addition**, search the item, enter the quantity (and lots if traced), and click **Add to batch**.
5. Record **in-process test** results in the lower grid; Pass/Fail is computed against the spec. Click **Save results**.

Recorded lines turn green with a checkmark. If a quantity is outside the recipe tolerance, or on-hand stock is short, an amber warning is shown — these are advisory and do not block you. The **Material variance** panel compares planned vs actual per line with cost variance and overall yield.

Express execution

To record everything at once, click **Express...** then **Record at standard**. This records every remaining material line at its planned quantity, consuming stock FIFO (lot-traced items take the oldest on-hand lots), and checks off all instructions. Any shortfalls are reported afterward.

Completing the batch

On a Released order, use the Lifecycle bar's **Complete** controls:

1. Enter the **Actual batch size**.
2. Enter a **Reason** if one is required.
3. Sign: if a signature is required, type **your password**, plus your **MFA code** if you are enrolled. Add a **witness** (email + password + optional note) if the "+ add witness" option is required or you choose to.
4. Click **Complete**.

Completing mints the finished-good stock for the produced lot and records the production receipt. If the actual batch size deviates from the planned size beyond the yield tolerance (default 5%), a warning like "Yield 95 deviates 5.3% from the planned 100 (tolerance 5%)" appears — the completion still succeeds. If your group is **not permitted to complete orders**, the form shows an amber notice and a **supervisor** can

authorize it in place by entering their own email, password, and MFA — their signature goes on the record. Completing requires the order-completion secured program; ask an administrator if you are blocked and no supervisor is available.

Once completed, the Lifecycle bar offers **Close order** (with an optional reason) to finalize it.

Reversing a completion

If a batch was completed in error, open the completed order and click **Reverse completion...** (available only on orders created in this system). Reversing returns the order to **Released**, removes the produced stock, restores the consumed materials, and resets the procedure for re-execution. It requires the same e-signature (reason / password / MFA / witness / supervisor elevation).

The system **refuses** the reversal when the produced stock is no longer untouched:

- **already consumed** by another batch;
- **already shipped**;
- its **QC test results were recorded**;
- its **QA release was dispositioned** (approved, held, etc.);
- it was **moved, split, or adjusted** (the message shows on-hand vs produced).

In each case, unwind the downstream action first (reverse the shipment, undo the disposition) before you can reverse the completion.

Specifying packouts

Click **Packouts** on a batch order to see how much bulk it makes, how much is already **allocated** to packaging orders, and how much **remains** (an amber "Over-allocated" badge warns if you commit more than the batch makes). While the batch is Not-started or Released, add a requirement:

1. Choose a **Packout** option.
2. Enter the **Make qty** (units to package).
3. Optionally cap the **Bulk to supply**; otherwise the full requirement is allocated.
4. Click **Create packaging order** — a linked packaging order is created and the bulk is allocated.

If the product has no packout options, the panel points you to **Item Update** → **Packaged Products**. On a packaging order, the same panel instead shows which batch order supplies its bulk.

Revising a released order

To change a released production order, open it and click **Revise order...** in the Revisions panel. This opens a **draft** and puts the order into **Being edited** — while a draft is open, execution and lifecycle actions are refused.

1. Edit line quantities/comments, **remove** or **restore** lines, or **+ Add line** (ingredient, instruction, or in-process test). Lines already executed are locked, and lines allocated to packouts cannot be removed.
2. Type a **Revision comment** (required) and click **Save comment**.

3. Under **Publish revision**, sign as configured (reason / password / MFA / witness, or supervisor elevation) and click **Publish revision**. The order returns to Released at the new revision number.

Click **Cancel draft** to discard the changes. Every published revision is kept in the **Revision history** with who published it and when.

Recording consumed lots (batch orders)

Two extra controls on a batch order record material consumption for traceability and costing:

- **Record consumed lots (specific)** — enter each raw-material **lot #** and quantity the batch used. This records lineage for recall, depletes each lot's on-hand by specific identification, and rolls the real cost into the produced lot's unit cost. If on-hand is short it is recorded anyway (flagged in amber). Refusals include "Lot X not found" and "A batch can't consume its own produced lot."
- **Consume by quantity (FIFO)** — for items that are not lot-traced, search the item and enter a quantity; stock is depleted oldest-first across lots. Lot-traced items are flagged "lot-traced → use specific" and steered to the specific path above.

Both show a result banner with the produced lot and its rolled-up unit cost.

Sales & shipping

This section covers selling and shipping finished goods: raising a shipping order, reserving stock, recording what actually left the dock, and producing the packing slip and invoice. Everything except the paperwork browsers lives on the **Orders** page (top navigation bar). Packing slips, invoices and price lists each have their own tab.

Create a shipping order

1. Open **Orders** and click **New shipping order** (top right).
2. In **Customer (bill-to)**, type a customer name or code and pick from the list. This is the account that gets billed.
3. Fill the optional header fields as needed: **Customer PO #**, **Reference**, **Terms** (a dropdown of your set-up terms codes) and **Carrier** (search by name/code).
4. Add lines: type in **Add item — search by code or description...** and click a match. Each line lands with quantity 1. Edit **Qty**, **Unit** and **Unit price** in place; the **Extended** column and the **Total** update as you type. Click **remove** to drop a line.
5. Click **Create shipping order**.

Price sourcing. When you add an item the app first looks up the customer's assigned price list and uses that price (respecting quantity-break tiers). If the customer has no list price for that item, it falls back to the item's own sale price. Because prices are customer-specific, **changing the customer clears all the lines** — set the customer first.

Return lines. A **negative quantity** is a return (a credit). A zero or blank quantity is simply ignored. You need at least one non-zero line and a customer to create the order.

The order is created **Not started** and its detail panel opens. Common refusals: "not flagged as a customer" means the entity you picked isn't marked as a bill-to — pick a real customer or have the entity flagged first.

Edit lines before releasing

While the order is **Not started**, click **Edit lines**. You can change each line's qty/unit/price (**Save** per line), **remove** a line, or use **Add a line** (the customer's list price is sourced automatically). A separate **Edit order** button lets you change the reference and required date. If your group is set up as request-only, saving shows "Submitted for approval (request #...)" and the change waits for an approver — it does not take effect immediately.

Move the order forward with the **Lifecycle** bar: **Release**, then **Complete** (this may prompt for an electronic signature — see below), then **Close order** with an optional reason.

Stage stock into a shipping assembly

Staging reserves specific on-hand parcels so nothing else can grab them before the truck loads. Click **Stage / reserve stock**.

1. Click **New assembly**. It appears with a location code and a green **open** chip.

2. In the assembly, choose a line from **Stage for line....** Free on-hand parcels appear as indigo chips (lot · quantity · location).
3. Click a chip to add an entry row, adjust the quantity (up to the "of ..." maximum), then click **Stage into assembly**.
4. Each staged parcel is listed with an **unstage** link to release it.

Reserved stock is untouchable by batch orders and other shipping orders. Only **lot-traced** items can be staged; a line marked "not lot-traced — not stageable" ships by quantity instead. If the order is already completed or closed you'll see an amber note — you can still **unstage**, but not stage new stock. Parcels marked "staged in legacy" were reserved in the old system and must be released there. Use the **Label** link on an assembly to print its assembly label, which doubles as the shipping label.

Record shipped lots

At shipping time, click **Record shipped lots**. For each lot-traced line you'll see chips to pick from:

- **Emerald "staged" chips** — stock you reserved in an assembly. These pre-fill the entry rows automatically and ship first.
- **Indigo chips** — other on-hand lots for that item.

Click a chip (or + **add lot manually** and type a lot) to add a row, enter the quantity, optionally set a **Ship date**, and click **Record shipped lots**. A **negative quantity is a customer return** — the lot comes back into stock and bills as a credit. Each time you record lots, a **packing slip** (shipment event) is created, so you can ship and invoice in partial batches. If the order has no lot-traced items you'll be told to enable lot tracking on those items first.

Shipments panel and reversing a shipment

The **Shipments** panel lists every packing slip on the order with its date and lot chips, and a **Packing slip N** link to view/print it. To undo a shipment, click **Reverse shipment....** Reversing restores the shipped stock where it left (back into the shipping assembly if it was staged), unwinds the shipped quantities, and stamps the slip **REVERSED**.

Two common refusals:

- "...already invoiced — reverse the invoice first" — you must reverse the covering invoice before the shipment.
- "...no longer at its restock location" — the returned or consigned stock has since moved, so it can't be cleanly restored.

Reversal requires an **electronic signature** whose exact prompts depend on configuration: a **reason**, **your password** (and **MFA code** if you're enrolled), an optional or required **witness** (email + password + MFA), and — if your group isn't permitted to reverse shipments — a **supervisor** email/password to authorize it (their signature goes on the record). This needs the *orders.reverse* program; ask an administrator if the button refuses outright.

Generate and reverse invoices

Once the order is past Not started, click **Generate invoice...**, optionally enter a **Freight charge**, and click **Generate invoice**. It bills only the shipped quantities no earlier invoice covered, applies taxes from the accounting tax rules, and gives you a link to the invoice. Because invoices are per shipment event, you can invoice after each partial shipment.

The **Invoices** tab lists every invoice (number, date, customer, order, PO, total). **View / print** opens the printable document. **reverse** posts a credit with the same number and negated lines, after a confirmation prompt; the shipped quantities then become invoiceable again. A reversed invoice prints with a red "Credit — reversal of invoice..." banner.

Packing slips

The **Packing Slips** tab lists all slips with a red **REVERSED** badge on any that were reversed. **View / print** shows Bill To / Ship To, order and PO numbers, carrier and quantities (ordered / shipped / back-ordered). Reversed slips print a large red **REVERSED** stamp so they can't be mistaken for live paper.

Price lists

Open **Price Lists** (needs the *sales.priceLists* program; editing needs *sales.priceListEditor* — a refusal appears as an error message). Click **New price list** (name, optional code). **Manage** a list to work three sections:

- **Versions** — add an effective-dated version; the one effective today is tagged "effective now". You can only price items once a version is effective.
- **Prices** — add a priced item with up to five **quantity-break tiers** (min qty → price), optional packaging (package type, qty per package, unit, and "prices are per package"), and the customer's own item code.
- **Customers on this list** — assign or unassign customers by code. Assigned customers automatically get these prices on new shipping orders.

Warehouse transfers (consigned stock) and transfer invoices

If a shipping order's ship-to is a **warehouse / consignment** location, recording shipped lots does **not** consume the stock — it **relocates** it into that destination's consigned stock, still owned by the warehouse. Consigned stock is protected: batches and other orders can't draw it. These orders **cannot take return lines** (a negative quantity is refused) — bring consigned stock back with an inventory transfer from the consigned location instead. Their invoice is a **transfer invoice**, a stock-movement document with lines at zero price and no taxes or freight; entering a freight charge on one is refused.

Purchasing & receiving

This section covers ordering materials from suppliers, receiving them into stock, and the records that result. The main screens live under **Purchasing**, **Supplier Pricing**, **Supplier Bills**, and **Inv. Receipts** in the left navigation.

Set up supplier pricing first (Supplier Pricing)

Prices you enter here flow automatically onto purchase-order lines, so it pays to keep them current. Open **Supplier Pricing** (titled "Supplier Price Versions"). Search a supplier by code, then click **Manage**.

A supplier's prices live inside a dated **version**. In the **Versions** area:

1. Pick an **Effective date** (defaults to today) and an optional **Comment**.
2. Click **Add version**. The version that is effective as of today shows an **effective now** badge.

If the panel says "No version is effective today," add one dated today before you can price anything.

In the **Prices** area, click into "Add a priced item":

1. Search and pick the **Item**; optionally enter **Their item code** (the supplier's own part number).
2. Enter one or more **Quantity-break tiers** as *min qty* → *price* (up to five, cheaper prices at higher quantities).
3. Optionally set a **Package type**, **Qty / package**, and **Package unit** — e.g. 25 kg per DRUM.
4. Tick **Prices are per package (not per unit)** for by-package pricing. With this on, the tier price is the price of a whole package, not one unit.
5. Optionally set a **Lead time** and pin a **Manufacturer**.
6. Click **Add price**.

Common refusals: "Enter at least one tier price" (no price filled in); "Enter a price for every tier that has a min qty" (a tier has a quantity but no price). Editing prices requires the price-version editor permission — if you only have view rights, saving fails with a permission error; ask an administrator.

Create a purchase order (Purchasing)

Open **Purchasing** and click **New purchase order**.

1. Search and select the **Supplier** by name or code. (Changing the supplier clears all lines, because pricing is supplier-specific.)
2. Optionally set **Required date**, **Reference**, **Terms** (dropdown), and **FOB / Incoterms**.
3. In "Add item," search by code or description and click an item. Its **Unit price** and packaging are pulled from the supplier's effective price version; a blue note under the description shows the pack size and "your code ... (from price version)."
4. Adjust **Qty**, **Unit**, or **Unit price** on any line; the **Extended** and **Total** update live. Use **remove** to drop a line.
5. Click **Create purchase order**. The PO is created **Not-started** and its printable copy opens automatically.

You need the create-purchase-order permission. If the chosen entity isn't flagged as a supplier you'll see "Entity ... is not flagged as a supplier."

Editing lines later: while a PO is still **Not-started**, an **Edit lines** link appears on its row. You can change qty, unit, price, add lines, or remove them. Some changes apply immediately; others return "Submitted for approval (request #...)" and only take effect once a qualified approver signs off (see **Line Approvals**). You can't reduce ordered qty below what's already received, remove a line that has receipts, or leave a PO with no lines.

Print the PO and the pickup slip

On any PO row:

- **View / print** opens the print-faithful Purchase Order — supplier, ship-to, terms, line table with **Price Per** and **Value**, the **Total**, and a second page of standard Terms & Conditions. Use the **Print** button.
- **Pickup** opens the **Pickup Copy** — identical but with all pricing removed (no Price Per, Value, or Total). Hand this to a driver collecting the goods.

Receive against a purchase order

On the PO row, click **Receiving**. The panel lists each line's **Ordered**, **Received**, and **Backordered** quantities (backordered shows amber until the line is filled).

Under "Receive a shipment," for each line:

1. Enter the **Qty** arriving (it defaults to the outstanding backorder).
2. Enter the **Manufacturer lot** number. This is **required** by default and is your recall key.
3. Optionally enter the number of **Containers**.
4. If one delivery arrived as several manufacturer lots, click **+ split into another lot** and enter each separately.
5. Optionally add a **packing-slip / receipt reference**, then click **Record receipt**.

What a receipt creates: every received lot is given a **new system lot number** and on-hand stock at the receiving location, tagged with the supplier and the manufacturer lot so it can be traced in a recall. The line's **Received** rises and **Backordered** falls, and the receipt appears under **Receipt history** (receipt number, date, our lot, mfr lot, qty, containers).

Refusals: "Enter a manufacturer lot number for each received lot" (a lot line has a quantity but no manufacturer lot, and the plant policy requires it); "Cannot receive against a closed purchase order." Receiving requires the receive permission. If an administrator has turned the manufacturer-lot policy off, that field becomes optional.

Recall lookup: the **Recall lookup** button at the top of Purchasing finds every received lot for a manufacturer lot number — showing our lot, supplier, quantity, cost, receipt date, the originating PO, and a **trace forward** link into the batches and shipments that used it.

Reverse a receipt (untouched-stock rule)

Receipt reversal is done from the **Inv. Receipts** page. On the receipt's row click **Reverse**, enter a **Reason** (required), and click **Confirm reversal** — this removes the on-hand the receipt created.

A reversal is only allowed while the received stock is **completely untouched**: exactly one parcel still holding the full received quantity. If any of it has been moved, split, consumed, adjusted, or already counted, you'll see "Cannot reverse — the received stock has since been moved, split, consumed, or adjusted." A receipt already reversed reports "This receipt has already been reversed." Reversal requires the reverse permission.

Miscellaneous receipts — receive without a PO (Inv. Receipts)

Use **Inv. Receipts** to bring stock in with no purchase order — opening balances, found stock, or samples. Click **New receipt**:

1. Search and add items; for each line enter **Qty**, **Unit**, an optional **Mfr lot**, optional **Unit cost**, and **Containers**.
2. Add an optional **Reason / reference** and click **Record receipt**.

Each line mints a new system lot and on-hand at the receiving location. The manufacturer lot is optional here — set it if you want the lot to be recall-findable. These receipts are reversed the same way (Reverse → Reason → Confirm), under the same untouched-stock rule. Creating misc receipts requires the receipts permission.

Supplier Bills

Supplier Bills is a register of supplier invoices already in the system. Search by invoice or bill number, then click **View / print** to open a bill: supplier, line items with quantities and amounts, plus subtotal, additional cost, tax, and total. Each bill's lines tie back to the received quantities and PO lines behind them, so a bill shows what was actually delivered and costed. This screen is for finding, reviewing, and printing bills — it does not key new bills. Viewing bills requires the bills permission.

Inventory & quality

This section covers day-to-day stock handling and QA release: browsing and adjusting inventory, counting a location, turning on lot tracking, tracing a lot for a recall, and signing off lot dispositions and certificates. The relevant menu items are **Inventory**, **Inv. Counts**, **Lot Tracking**, **Lot Trace**, **Certificates**, **Item Tests**, and **QA Approvals**.

Inventory — browse, adjust, move, and label parcels

Open **Inventory**. Each row is one *parcel* (a physical container of stock): item, description, lot, subplot, location, quantity, and QA status.

1. Filter with the **Item code** box, or tick **On-hand only** to hide empty parcels.
2. Click a **Qty** or **Status** header to sort.
3. Each row has three actions on the right: **Label**, **Adjust**, and **Move**.

Adjust (write stock up or down to a counted figure):

1. Click **Adjust**. The dialog shows the item, lot, location, and current quantity.
2. Enter the **Counted quantity** (zero or more) and a **Reason** — the reason is mandatory (for example "cycle count", "spillage", "correction").
3. Click **Apply adjustment**. The system records the change under a COUNT change set and writes an audit entry. If the difference exceeds the configured tolerance, a *reweigh outside threshold* notification is raised for review.

Move (relocate stock between locations):

1. Click **Move** (disabled when the parcel is empty).
2. Enter the **Quantity to move** — it must be positive and no more than what is on hand. Any remainder stays at the source location.
3. Type in **Destination location** to search, then pick a location from the list.
4. A **Reason** is optional. Click **Move stock**. The moved quantity merges into (or creates) a parcel at the destination.

Label opens a printable container/lot label — item, our lot, manufacturer and supplier lots, quantity, dates, and QA status — with a **Print** button. Reprinting is just reopening this from the row.

Protected stock (common refusals). Some parcels cannot be adjusted or moved directly:

- *"This parcel is reserved to a shipping order — unstage it from the order's staging panel instead."* The stock is staged to an order; release it there first.
- *"Sample parcels are managed by the QA sampling flow — not by plain adjustment."* Retained QA samples.
- *"Assembly parcels are managed from the shipping order's staging panel." / "Assemblies are filled from the shipping order's staging panel — not by plain transfer."*

You also cannot adjust to a negative number, or move to the same location. Browsing needs the *inventory browser* program; adjusting and moving need *inventory adjust* and *inventory transfer* respectively — ask

an administrator if the buttons refuse you.

Inventory Counts — count a location and post one change set

Open **Inv. Counts** to run a formal count sheet that applies every correction under a single change set.

1. Click **New count**. Choose a **Location to count** (required). Optionally narrow to one **Item**, add a **Status filter** (e.g. "Approved"), and a **Description**.
2. Click **Create count**. The sheet snapshots every on-hand parcel at that location at that moment — its list of lines will not change afterward.
3. In the count panel, type the **Counted** figure for each line. The **Adjust** column previews the drift live (green for a gain, red for a loss).
4. Click **Save counts** to store your figures (you can leave and return; drafts stay editable).
5. Click **Post count**. Every non-zero difference is applied under one COUNT change set, and the panel shows the change-set number. Posted counts are read-only.

Notes and refusals: you must **Save** before **Post** (you will see "Save your counts before posting"). Posting needs at least one entered figure. If nothing was on hand when the sheet was created, it reads "No parcels were on hand at that location when the count was created." Sample (SMP) and assembly (ASM) locations cannot be counted this way. **Delete draft** discards an unposted sheet; a posted count cannot be deleted. Requires the *inventory count* program.

Lot Tracking — turn on lot tracing per item

Open **Lot Tracking**. Items show as **Lot-traced** or **FIFO (not traced)**. Until an item is traced, it is consumed first-in-first-out by quantity only.

To enable an item, click **Enable...** and enter its opening on-hand stock by lot:

1. Choose a **Location / warehouse**.
2. For each lot, pick a type: **Raw (vendor lot)** — enter the vendor/manufacturer lot and ERP1 assigns its own lot number — or **Finished good** — enter the existing lot number. Add the **Qty on hand** and optional **unit cost**.
3. Use **+ add lot** for more lots, or **+ add another location** for stock in more than one place.
4. Click **Enable lot tracking**.

Important: enabling *replaces* the item's current on-hand inventory with exactly the lots you enter (shown as an amber warning). Afterward a table lists the ERP1 lot numbers to write on the physical containers.

Disable turns tracing back off. Requires the *lot tracking* program.

Lot Trace — trace a lot forward and back (recall)

Open **Lot Trace**. Type any identifier — a finished-good lot, a batch or packout lot, an ERP1 raw-material lot, or a supplier's manufacturer lot off the drum — and click **Run recall**.

The results show: the focus lot with its QA badge; **Made from** (source lots and declared ingredients); **Packed out as** (the packout lots); summary tiles; **On-hand inventory** for every form of the lot; and **Shipments containing the affected lots** — the recall list with customer, PO number, ship date, order and

quantity. If you searched a supplier lot, a banner shows how it matched to the ERP1 lot. **Reversed (cancelled) shipments are excluded** — only genuine outbound shipments appear, captured when the shipping order is closed. Expand **Data coverage & limitations** for caveats. No match shows an amber hint to try the ERP1 lot or the manufacturer lot. Requires the *recall* program.

QA — test results and dispositions

When a batch completes, ERP1 creates a sample set for the lot: the lot starts on **Hold** with a retained sample. Quality staff then record results and set the final disposition. This is done from the lot in **Lot Trace**.

Enter test results: on the focus lot, click **Enter test results**. Each row is a test with its specification; type the result and **Save results**. Pass/Fail is computed automatically against the spec — you do not set it. (The tests and specs themselves are configured on the **Item Tests** page under *Test requirements*; that page also drives the Certificate of Analysis. Editing tests needs the *item tests edit* right.)

Change disposition: click **Change disposition**, choose **Approved**, **Hold**, or **Rejected**, and optionally set **Grade** and **Expiry**. A disposition is electronically signed:

- Enter a **Reason** if the item marks it required.
- Enter **your password** to sign, plus an **MFA code** if you use two-factor.
- Add a **witness** (email, password, MFA) when the item requires one, or via **+ add witness**.
- If your group is not permitted to disposition, an amber note appears and a **supervisor** can authorize it in place (their email, password, and MFA — their signature goes on the record).

If your group may only *request* dispositions, submitting returns "Submitted for approval" with a request number; it takes effect once an approver acts. Requires the *disposition* program; entering results requires the *results* program.

QA Approvals and Certificates

QA Approvals lists pending disposition requests (lot, item, what was requested, by whom, when). Click **Approve** to enact it — signing with your password/MFA (and witness, if required) — or **Reject** with a mandatory reason. Approving needs an approving capability for your group.

Certificates lists every Certificate of Analysis by product, lot, grade, disposition, and manufacturing date. Search by product or lot, then click **View / print** to open the printable CofA in a new tab. Requires the *cofa* program.

Planning, viewers & accounting

These four pages — **Planning**, **Viewers**, **Notifications** and **Accounting**, plus the **Dashboard** you land on at sign-in — are the office-side reporting and setup tools. They mostly read and summarize; the few buttons that change things (recalculating the plan, creating a purchase order, running an export) are called out below.

Dashboard

The Dashboard opens automatically when you sign in (the **Dashboard** tab in the top bar returns you here).

- The row of tiles across the top — Entities, Items, Recipes, Orders, Lots, On-hand containers, Lineage edges, Audit records — are live counts. Click any tile to jump to that page.
- **Orders by type** breaks the order count into Batch, Packaging, Purchase and Shipping.
- **Recent activity** lists the latest changes with who made them; **View audit trail** opens the full history.
- The three cards at the bottom show System status, the Last import (status, mode and when it finished) and the running Version / database / cache health. If System status is not "ok" or the last import failed, tell an administrator before relying on the numbers.

Planning

The **Planning** page has three tabs — **Plan Tracing**, **Short Inventory** and **Supply & Demand** — and one action button, **Recalculate Plan Trace**.

Reading the plan (Plan Tracing). This grid lists every requirement the plan found. The **Reference** column tells you how each requirement is being filled, in the order the planner tries to cover it:

1. **AVAILABLE** (or a consignment area's name) — covered from available stock on hand.
2. **Hold / Expired** — covered from quarantined or expired stock (flagged so you can check it).
3. **MF#<number>** — covered by an open batch/manufacturing order that will produce it.
4. **PO#<number>** — covered by an open purchase order that will receive it.
5. **Short** (shown in red) — nothing covers it; a purchase order must be raised.
6. **Negative** — a minimum-stock shortfall.

Use the **Reference** dropdown to filter to one kind, the search box to find an item, and the column headers to sort. A note above the grid says when the plan was last recalculated and whether you are looking at the native ERP1 plan or the legacy nightly plan.

Recalculating. Press **Recalculate Plan Trace** to rebuild the plan from current data. You'll be asked to confirm ("the viewers will switch to the native plan"). When it finishes, a green line summarizes how many requirements, short lines and demands were produced. This requires the plan-recalculate program; if the button does nothing, ask an administrator for access.

Creating a purchase order from Short lines. On the Plan Tracing tab, a checkbox appears only on rows that can be ordered — **Short** or **Negative** lines that don't pin a specific lot. Tick the lines you want, then press **Create purchase order (n)**. All ticked lines must be the **same item and the same required manufacturer**; the quantities are summed into one order. Common refusals:

- *"only Short/Negative requirements can be ordered"* — you selected a covered line.
- *"requires a specific subplot — a purchase order cannot supply it"* — that requirement is pinned to a lot.
- *"All selected lines must be for the same Item and Required Manufacturer combination."*
- *"No supplier pricing is set up for ... — add a price version detail first"* — set up the supplier's price on the Supplier Pricing page, then retry.

If more than one supplier prices the item, a bar appears: **"Several suppliers price this item — which one?"** Pick a supplier (the preferred one and its price are shown), then **Create with this supplier**, or **Cancel**. On success a green line gives the new PO number, supplier, quantity and item.

Short Inventory is a plain summary of everything that can't be covered — one line per item / required manufacturer / required lot, with the short quantity, quantity on hand, the date it's needed, an "order by" date and the suggested supplier. If it says "Nothing is short," every requirement is covered.

Supply & Demand is read-only. Search for and pick an item; the page then shows two tables — **Sources** (warehouse stock, purchases, batches, packaging) and **All demand** — plus a totals bar (available stock, held/rejected, total supply, open demand, balance). Click a source or a demand row to grey out everything except the lines it's allocated to.

Viewers

The **Viewers** page is a set of filterable, exportable report grids. You'll see cards for each report you're allowed to open (if none appear, your role doesn't include them). The nine reports are: **Shipment Detail**, **Open Shipping Order Detail**, **Open MF Order Detail**, **Inventory Movement**, **Purchase History**, **Batching Orders**, **Where Used**, **Inventory At Date**, and **Complete MF Orders**.

Open any card and you get the same controls:

1. Fill in the **filters** at the top — date ranges, dropdowns (e.g. movement type, order status) or a text box — then press **Apply**. Filters marked with a red asterisk are required; *Inventory At Date* needs an "As of" date and won't load until you set one ("Set the required filters and press Apply").
2. Sort by clicking a column header; use the search box for a quick text find; page through results at the bottom.
3. Press **Export** to download the whole result set (not just the page you're viewing) as a CSV file. The export uses the same filters as the grid, so set and **Apply** any required ones first or you'll get an "Export failed" message.

Use **All viewers** (top right) to return to the list.

Notifications

The **Notifications** page controls automatic e-mails and has three cards.

Notification rules lists each configured rule by code and security group. The **Status** badge reads **Active** (ERP1 sends that e-mail) or **Not fired by ERP1** (kept for parity/future use). Click a row to edit it: set the **Security group** (* = any group), the **Send to** address list (separate multiple addresses with semicolons), tick **Use send-to list only** to suppress automatic recipients, and edit the **Subject** and **Text**. The Text is an e-mail template — the **Insert field** buttons drop in @Field placeholders that get filled in when the event happens. **Save rule** commits. You can also add per-area send-to additions, or **delete** a rule. To add one,

pick a code from the dropdown at the bottom, enter an address and press **Add rule**.

E-mail log shows every e-mail ERP1 has queued, filterable by status (Not sent / Sending / Sent / Failed). Click a row to preview the rendered message. **Process queue now** attempts delivery immediately; failed items retry up to five times, and a **requeue** link re-sends a failed one. Legacy history rows (shown with an "L" number) are never sent.

Mail settings holds the SMTP server details and switches. Delivery stays off until `notifications.enabled` is set to true — until then e-mails just accumulate in the log as a dry-run trail. Enter the server, port, user, password and "from" address, press **Save settings**, then use **Send a test e-mail to** with **Send test** to confirm it works. Editing these requires the Configuration (admin) program; if the fields won't load, ask an administrator.

Accounting

The **Accounting** page defines how items post to your external accounting system, and exports the journal.

- **GL groups** — every item belongs to a GL group. Click a group to open its **Account mappings** grid on the right, where each GL code (Asset, COGS, Income, adjustment reasons) is pointed at an external account. Add a group with the form at the bottom; add or remove mappings in the grid.
- **GL codes** and **Account codes** are simple code/description lists you add to or delete from. Account codes must match names in your accounting system exactly.
- **Tax rules** apply per level (1 = federal, 2 = state/provincial, 3 = municipal). Each rule matches a customer tax group and an item tax group (a blank item group is the customer's default) and sets a rate and/or per-unit amount, with an optional **Tax on tax**. Add, **edit** or **delete** rules in the table. Use **Tax preview** to dry-run a customer and hypothetical lines and see the computed tax per level.

Accounting export produces a journal for a date range. Set **From** and **To**, choose the **Format** (QuickBooks IIF file or CSV journal), and tick which kinds to include — Invoices, Purchase receipts, Misc receipts, Adjustments, Builds. Press **Preview** to see the entry counts per kind and any warnings (for example, an item whose GL group has no account — those land on a fallback account). Press **Download** to save the file; each run is logged in the **Export history** table with the range, format, entry count and who ran it. In short, the export books invoices to receivables/income/tax, purchase receipts to payables/asset, and the native misc receipts, adjustments and builds to their mapped accounts.